GATE-AII BRANCHES
Engineering Mathematics



SINGLE VARIABLE CALCULUS

Lecture No.- 11



By- Rahul sir

Recap of previous lecture



Topic Rolles theorem

Topic Lagrange mean value theorem

Topic Problems based on mean value theorem

Topic Basic integration

Topics to be covered



- Topic: Integration via parts
- Topic: Definite integrals
- Topic: Improper integrals
- Topic: Area bounded regions

✓ Calculus
Single variable
CSE

✓ Linear Algebra
Recorded

Calculus - 12
✓ Taylor SERIES
✓ Partial derivatives

✓ Jan

{ ME | CE | EE | EC | EI | PI

CSE {
- Prob
- CALCULUS]
- ✓ Linear algebra - CS ✓ free
- ✓ Multivariable Calculus
- ✓ Vector Calculus
- ✓ Numerical methods
- ✓ Transform Theory
- ✓ O.E
- ✓ P.D.E
- ✓ complex Analysis
- ✓ fourier SERIES

2 marks
EE/EC/EI
ME/CE/CS

Improper Integral:

- Improper Integral of First Kind
- Improper Integral of SECOND Kind

Improper Integral of First Kind:

$\int_a^b f(x)\,dx$ "Range of Interval is Infinte"

(A) $\int_a^{\infty} f(x)\,dx = \lim_{t\to\infty} \int_a^t f(x)\,dx$ → finte Convergent / Infinte Divergent

(B) $\int_{-\infty}^{a} f(x)\,dx = \lim_{t\to-\infty} \int_t^a f(x)\,dx$ → Finte Convergent / Infinte Divergent

(C) $\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{0} f(x)\,dx + \int_0^{\infty} f(x)\,dx = \lim_{t\to-\infty} \int_t^0 f(x)\,dx + \lim_{t\to\infty} \int_0^t f(x)\,dx$ → C finte / D Infinte

$$I_1 = \int_1^{\infty} \frac{1}{x}\,dx$$

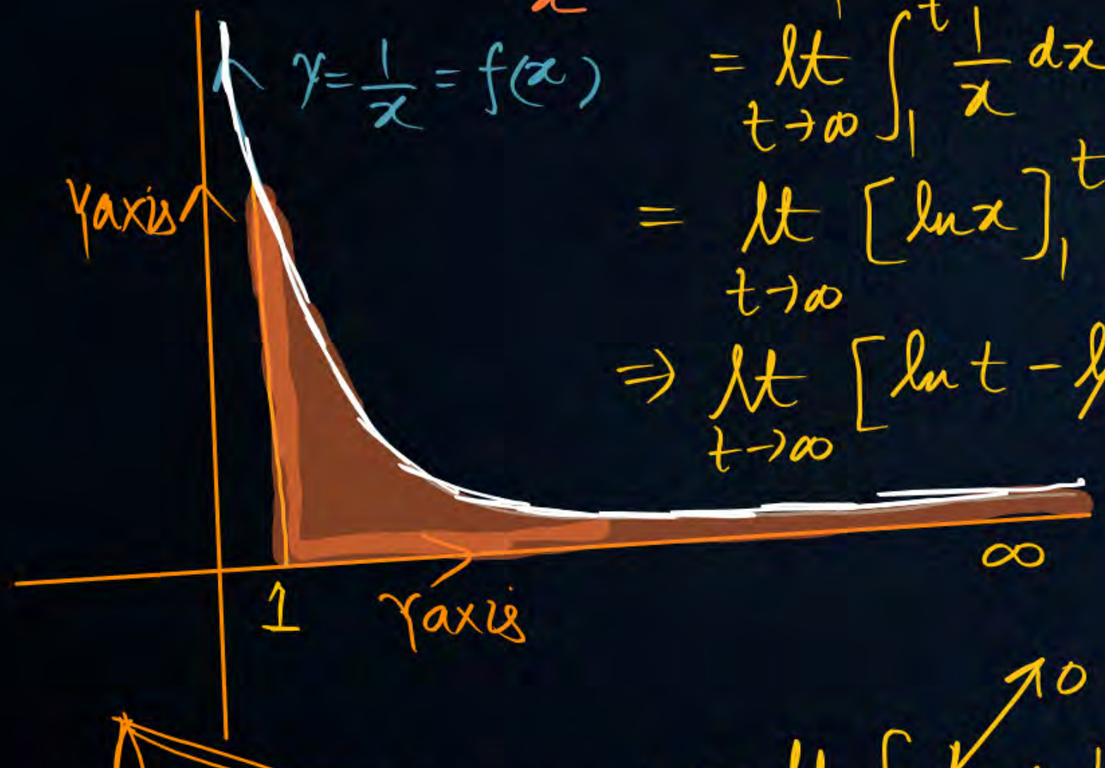
$$f(x) = \frac{1}{x}$$

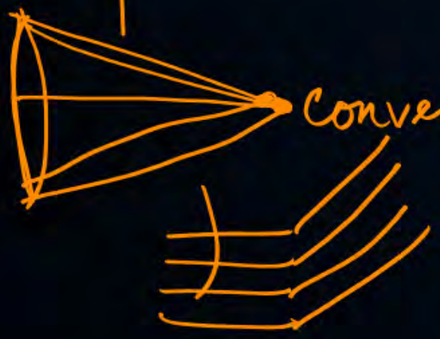
$$I = \int_1^{\infty} \frac{1}{x}\,dx = \text{(D)}$$

$$= \lim_{t\to\infty} \int_1^{t} \frac{1}{x}\,dx$$

$$= \lim_{t\to\infty} [\ln x]_1^{t}$$

$$\Rightarrow \lim_{t\to\infty} [\ln t - \ln 1] = \text{Divergent}$$





$$= \lim_{t\to\infty} \left[\frac{-1}{2t^2} + \frac{1}{2}\right] = \frac{1}{2}$$

Number — convergent

$$I = \int_1^{\infty} \frac{1}{x^3}\,dx$$

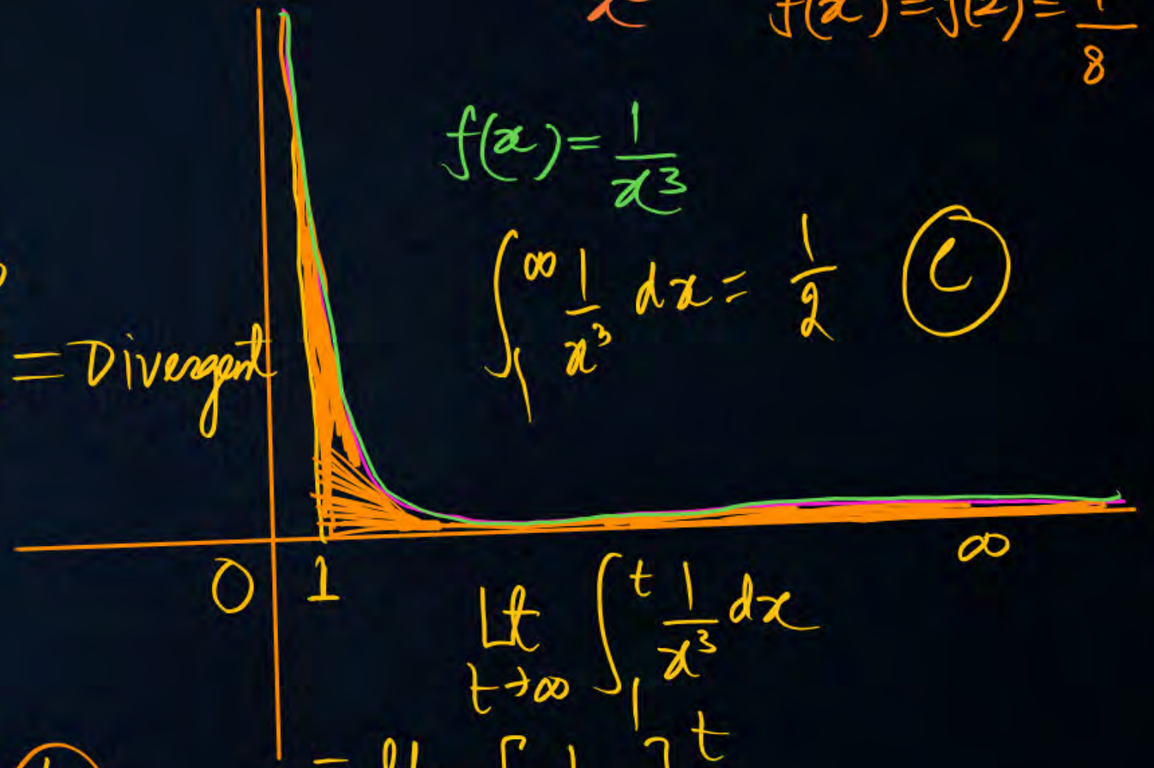
$$f(x) = \frac{1}{x^3}$$

$x = 2$

$f(x) = f(2) = \frac{1}{8}$

$$f(x) = \frac{1}{x^3}$$

$$\int_1^{\infty} \frac{1}{x^3}\,dx = \frac{1}{2} \quad \text{(C)}$$



$$\lim_{t\to\infty} \int_1^{t} \frac{1}{x^3}\,dx$$

$$= \lim_{t\to\infty} \left[\frac{-1}{2x^2}\right]_1^{t}$$

Type-02 Improper Integral of SECOND Kind :

$$I = \int_0^4 \frac{dx}{\sqrt{x}}$$

$$f(x) = \frac{1}{\sqrt{x}}$$

$$f(0) = \overrightarrow{\infty}$$

$$I = \int_0^{\frac{\pi}{2}} \tan x \, dx \qquad f(x) = \tan x \qquad f\left(\frac{\pi}{2}\right) = \tan\frac{\pi}{2} = \underline{\to\infty}$$

Improper Integral (SECOND Kind)

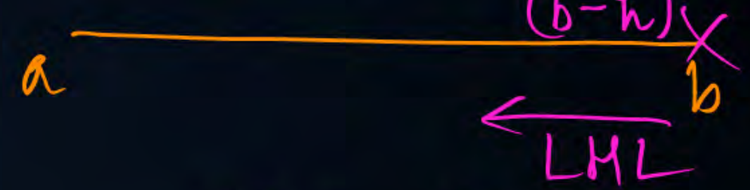
(A) If $x = a$ function is Not defined

(A) $$\int_a^b f(x)\,dx = \lim_{h\to 0} \int_{a+h}^{b} f(x)\,dx$$



(B) $\int_a^b f(x)\,dx \Rightarrow$ At $x = b$

$\lim_{h\to 0} \int_a^{(b-h)} f(x)\,dx$



Ex(1) $I = \int_0^4 \frac{dx}{\sqrt{x}} = \lim_{h\to 0} \int_{0+h}^4 \frac{dx}{\sqrt{x}}$

$= \lim_{h\to 0} \left[2\sqrt{x}\right]_{0+h}^4$

$= \lim_{h\to 0} \left[2\sqrt{4} - 2\sqrt{h}\right]$

$= (4) =$ Convergent

2) $I = \int_0^{\pi/2} \tan x\,dx$

$= \lim_{h\to 0} \int_0^{\pi/2-h} \tan x\,dx$

$= \lim_{h\to 0} \left[\log \sec x\right]_0^{\pi/2-h}$

$=$ Divergent



Topic : Single Variable Calculus

H.W

#Q. The value of $\int_{\sqrt{\ln 2}}^{\sqrt{\ln 3}} \frac{x\sin x^2}{\sin x^2 + \sin(\ln 6 - x^2)}\,dx$ is

A $\frac{1}{4}\ln\frac{3}{2}$

B $\frac{1}{2}\ln\frac{3}{2}$

C $\ln\frac{3}{2}$

D $\frac{1}{6}\ln\frac{3}{2}$



Topic : Single Variable Calculus

✓ 3-4 times

#Q. If for non zero x, $af(x) + bf\left(\frac{1}{x}\right) = \frac{1}{x} - 5$

where $a \neq b$, then $\int_1^2 f(x)dx = \ldots\ldots\ldots$

$$a f(x) + b f\left(\frac{1}{x}\right) = \frac{1}{x} - 5 \quad \text{— ①}$$

$x \longrightarrow \frac{1}{x}$

New Equation

$$a f\left(\frac{1}{x}\right) + b f(x) = \frac{1}{x} - 5 \quad \text{— ②}$$

$$a f(x) + b f\left(\frac{1}{x}\right) = \frac{1}{x} - 5 \quad \times a \quad \text{— ①}$$

$$a f\left(\frac{1}{x}\right) + b f(x) = x - 5 \quad \times b \quad \text{— ②}$$

Then $\int_1^2 f(x)\,dx$

$$a^2 f(x) + ab f\left(\frac{1}{x}\right) = \frac{a}{x} - 5a$$

$$ab f\left(\frac{1}{x}\right) + b^2 f(x) = bx - 5b$$

$$(a^2 - b^2) f(x) = \frac{a}{x} - 5a - bx + 5b$$

$$f(x) = \frac{1}{(a^2 - b^2)} \left[\frac{a}{x} - 5a - bx + 5b\right]$$

$$2x + 3y = 6 \times 6$$

$$3x + 6y = 12 \times 3$$

$18y$

$18y$

$$f(x)=\frac{1}{(a^2-b^2)}\left[\frac{a}{x}-5a-bx+5b\right)$$

$$\int_1^2 f(x)\,dx=\int_1^2 \frac{1}{(a^2-b^2)}\left[\frac{a}{x}-5a-bx+5b\right]$$

$$=\frac{1}{(a^2-b^2)}\left[a\ln x-5ax-\frac{bx^2}{2}+5bx\right]_1^2$$

$$=\frac{1}{(a^2-b^2)}\left[(a\ln 2-10a-2b+10b)-\left(a\ln 1-5a-\frac{b}{2}+5b\right)\right]$$

solve It

$$=\frac{1}{(a^2-b^2)}\left[a\ln 2-10a-2b+10b+5a+\frac{b}{2}-5b\right]$$



Topic : Single Variable Calculus

#Q. Which of the following integrals is unbounded?

Convergent = Bounded

Divergent = Unbounded

Convergent - bounded

A ✗ $\int_0^{\pi/4} \tan x\,dx = [\log \sec x]_0^{\frac{\pi}{4}}$

$= \log \sec \frac{\pi}{4} - \log \sec 0$

$= \log \sqrt{2}$

B $\int_0^{\infty} \frac{1}{1+x^2}\,dx = \lim_{t\to\infty} \int_0^t \frac{1}{1+x^2}\,dx$

$= \lim_{t\to\infty} [\tan^{-1} x]_0^t$

$= \lim_{t\to\infty} [\tan^{-1} t - \tan^{-1} 0]$

$= \frac{\pi}{2}$ = bounded

C $\int_0^{\infty} x.e^{-x}\,dx$

$= \lim_{t\to\infty} \int_0^t xe^{-x}\,dx$

= finite

= bounded

D $\int_0^1 \frac{1}{1-x}\,dx$

Unbounded

$$\int_0^1 \frac{1}{1-x}$$

$$f(x) = \frac{1}{1-x}$$

$$f(1) = \frac{1}{1-1} = \to\infty$$

✓ Improper second kind

$$= \lim_{h\to 0} \int_0^{1-h} \frac{1}{(1-x)}\,dx$$

$$= \lim_{h\to 0} \left[-\ln(1-x)\right]_0^{1-h}$$

$$= \lim_{h\to 0} \left[-\ln(1-(1-h)] + \ln 1\right]$$

$$= \underline{\text{Infinity}} = \underline{\text{Unbounded}}$$

$$= \lim_{h\to 0} \left[-\ln(1-1+h) + 0\right]$$

$$= \lim_{h\to 0} \left[-\ln h\right] = -\infty$$ Unbounded



Topic : Single Variable Calculus

$$I = \int_{-\infty}^{\infty} \frac{dx}{1+x^2}$$

#Q. The value of the integral $\int_{-\alpha}^{\alpha} \frac{dx}{1+x^2}$ is

A $-\pi$

B $\frac{-\pi}{2}$

C $\frac{\pi}{2}$

D π

$$I = \int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \int_{-\infty}^{0} \frac{dx}{1+x^2} + \int_{0}^{\infty} \frac{dx}{1+x^2}$$

$$= \lim_{t \to -\infty} \int_{t}^{0} \frac{dx}{1+x^2} + \lim_{t \to \infty} \int_{0}^{t} \frac{dx}{1+x^2}$$

$$\Rightarrow \lim_{t \to -\infty} \left[\tan^{-1} x\right]_{t}^{0} + \lim_{t \to \infty} \left[\tan^{-1} x\right]_{0}^{t}$$

$$\Rightarrow \lim_{t \to -\infty} \left[\tan^{-1} 0 - \tan^{-1} t\right] + \lim_{t \to \infty} \left[\tan^{-1} t - \tan^{-1} 0\right]$$

$$\Rightarrow \tan^{-1} 0 - \tan^{-1}(-\infty) + \tan^{-1}(\infty) - \tan^{-1}(0)$$

$$\Rightarrow 0 - \left(\frac{-\pi}{2}\right) + \frac{\pi}{2} - 0$$

$$= \frac{\pi}{2} + \frac{\pi}{2} = \boxed{\pi}$$



Topic : Single Variable Calculus

1 Hrs.

#Q. The integral $\int_0^1 \frac{dx}{\sqrt{(1-x)}}$ is equal to 2.

$$I = \int_0^1 \frac{dx}{\sqrt{1-x}}$$

$$\lim_{h\to 0}\left[-2\sqrt{1-x}\right]_0^{1-h}$$

$$\int_0^1 \frac{dx}{\sqrt{1-x}}$$

$$f(x) = \frac{1}{\sqrt{1-x}}$$

$$f(1) = \to\infty$$

$$= \lim_{h\to 0}\int_0^{1-h} \frac{dx}{\sqrt{1-x}}$$

$$\lim_{h\to 0}\left[-2\sqrt{1-x}\right]_0^{1-h}$$

$$= \lim_{h\to 0}\left[-2\sqrt{1-1+h} + 2\sqrt{1-0}\right]$$

$$= 2 \quad \text{Ans.}$$

$= \checkmark$



Topic : Single Variable Calculus

H.W

#Q. If $f(x) = R\sin(\frac{\pi x}{2}) + S$, $f'(1/2) = \sqrt{2}$ and $\int_0^1 f(x)dx = 2R/\pi$, then the constants R and S are, respectively

A $2/\pi$ and $16/\pi$

B $2/\pi$ and π

C $4/\pi$ and 0

D $4/\pi$ and $16/\pi$



Topic : Single Variable Calculus

#Q. Let f be a real valued function of a real variable defined as $f(x) = x - [x]$ where [x] denotes the largest less than or equal to x . The value of $\int_{0.25}^{1.25} f(x)dx$ is _____ (up to 2 decimal places).

$f(x) = x - [x]$

$$\int_{0.25}^{1.25} (x - [x])\, dx$$

$$= \int_{0.25}^{1} (x-0)\, dx + \int_{1}^{1.25} (x-1)\, dx = \left(\frac{x^2}{2}\right)_{0.25}^{1} + \left(\frac{x^2}{2} - x\right)_{1}^{1.25} =$$

$=$

$$= \int_{0.25}^{1} x\,dx + \int_{1}^{1.25} (x-1)\,dx$$

$$I_1 = \left[\frac{x^2}{2}\right]_{0.25}^{1} = \frac{1}{2} - \frac{(0.25)^2}{2} =$$

$$I_2 = \left[\frac{x^2}{2} - x\right]_{1}^{1.25} = \left(\frac{(1.25)^2}{2} - 1.25\right) - \left(\frac{1}{2} - 1\right) =$$

0.5 Ans



2 mins Summary

- Topic One
- Topic Two
- Topic Three
- Topic Four
- Topic Five

✓ Improper Integral
✓ Question

THANK - YOU